

Jiale Chen

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[Home](#) | [GitHub](#) | [Google Scholar](#) | [ORCID](#)

EDUCATION

- **Beijing University of Technology** 2023 – 2027
B.Eng. in Electronic Science and Technology (Honors Class) Beijing, China
 - Cumulative GPA: **3.76/4**; Junior-Year Weighted Average: **93.12/100**.
 - Selected Coursework: **Digital Integrated Circuit Design (95)**, **RF Integrated Circuit Design (97)**, **Microcontroller Systems (97)**, **Control Systems (99)**, **Deep Learning (98)**, **Advanced Mathematics (100)**.

RESEARCH EXPERIENCE

- **FPGA-Based Remote Laboratory Platform** Mar 2025 – Aug 2025
Project Leader, Advisor: Prof. Sujuan Liu | Beijing University of Technology [🔗]
 - Designed and implemented a **remote FPGA laboratory platform** integrating **FPGA, Raspberry Pi 5, and STM32** for end-to-end remote hardware experimentation and debugging.
 - Developed a web-based control system supporting **remote JTAG programming, logic analysis**, virtual oscilloscope functions, digital-twin visualization, and real-time camera monitoring.
 - Implemented **UART-based FPGA bitstream transfer at 115200 baud**, achieving **10–20 s remote deployment latency** and reliable hardware configuration.
 - Built a **real-time signal acquisition and debugging framework** using **FPGA ADC/DAC modules** and protocol-level analysis for waveform visualization and system validation.
- **Efficient Adaptation and Deployment of Multimodal Large Language Models** Mar 2026 – Present
Core Member, Advisor: Hongyu Zhao | Shanghai Jiao Tong University
 - Developed **MedFusion**, a **shared LoRA adaptation framework** for multi-source multimodal learning using fused image-query representations for adapter grouping and dynamic routing.
 - Investigated **parameter-efficient adaptation, low-bit quantization**, and **memory-efficient fine-tuning** for multimodal large language models, with emphasis on **efficient inference and deployment**.
- **Communication-Efficient Model Compression for FedDyn in Federated Learning** Dec 2025 – Mar 2026
Core Member, Advisor: Prof. Ruhui Ma | Shanghai Jiao Tong University [🔗]
 - Investigated **communication-efficiency bottlenecks** in FedDyn under heterogeneous federated learning settings and analyzed the trade-off between **transmitted parameter volume** and convergence stability.
 - Developed a **FedDyn-aware parameter-importance strategy** for selective communication, reducing transmission overhead while maintaining model performance.

INDUSTRY EXPERIENCE

- **Glenfly** Jun 2026 – Aug 2026
Software Testing Intern Shanghai, China
 - Conducted **functional, stress, stability, and compatibility testing** on **Glenfly GPUs** across Windows and Linux, covering **OpenGL, OpenCL, Direct3D**, and system-level workloads.
 - Performed repeated **GPU workload validation and failure reproduction** to evaluate driver compatibility, system stability, and platform behavior under intensive workloads.
 - Developed an automated desktop-folder detection pipeline based on **DexiNed and SAM** to support repetitive validation workflows across **1080p, 2K, and 4K** displays with different DPI settings.

• Cisco [🌐]

Technical Engineer Intern

Jan 2026 – Feb 2026

Beijing, China

- Investigated **AI data-center networking for distributed GPU training**, focusing on **RDMA, RoCEv2, ECN/PFC**, and **Leaf-Spine** network architectures.
- Analyzed **switch architectures and high-performance network designs**, and supported validation of network switches and optical modules under data-center scenarios.
- Configured and validated enterprise routing and switching environments using **BGP, OSPF, VLAN, STP, and ACL** for protocol verification and troubleshooting.

• Lenovo [🌐][🔗]

Embedded Algorithm Engineer Intern

Jun 2025 – Aug 2025

Beijing, China

- Developed **Stable Diffusion 3 (SD3) inference pipelines** for text-to-image, image-to-image, and inpainting, integrating **SAM/SAM2** for segmentation and mask-guided generation.
- Built a **large-scale data-processing pipeline** for millions of image samples and trained **ControlNet** on image-mask-prompt triplets for segmentation-guided controllable generation.
- Converted the **SD3 + ControlNet** inference pipeline from **PyTorch to OpenVINO** and deployed it on **Intel GPUs**, completing end-to-end inference validation and **hardware-accelerated deployment**.

PUBLICATIONS & MANUSCRIPTS

- [1] Jiale Chen.
Communication-Efficient Compression Method for FedDyn Federated Learning Models. [🔗]
IEEE 4th International Conference on Control, Electronics and Computer Technology (ICCECT), 2026.
Published in IEEE Xplore.
- [2] Hongyu Zhao*, Jiale Chen*, Yihong Chen*.
MedFusion: Shared LoRA Adaptation for Multi-Source Medical Image Classification.
Manuscript in preparation for submission to *IEEE Transactions on Multimedia (TMM)*, 2026.

* Equal contribution.

HONORS & AWARDS

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| • Second Prize, National Undergraduate IC Innovation and Entrepreneurship Competition
<i>Talent Exchange Center, Ministry of Industry and Information Technology (MIIT), China</i> | Aug 2025
[🌐] |
| • First Prize, Beijing Integrated Circuit Design Competition (Digital Track)
<i>Beijing Municipal Education Commission</i> | May 2026 |
| • Third Prize, Chinese Collegiate Computing Competition (National Final)
<i>Organizing Committee of the Chinese Collegiate Computing Competition</i> | Jul 2026
[🌐] |
| • Second Prize, China Undergraduate Mathematical Contest in Modeling (Beijing Region)
<i>China Society for Industrial and Applied Mathematics (CSIAM)</i> | Nov 2025
[🌐] |

TECHNICAL SKILLS

- **Programming:** C, C++, CUDA, Python.
- **Hardware & EDA:** Verilog, Vivado, Quartus II, ModelSim, Cadence Virtuoso, STM32CubeMX.
- **AI & Deployment:** PyTorch, Hugging Face Transformers, PEFT, OpenVINO, OpenCV.
- **Systems & Development:** Linux, Git, Slurm.
- **English:** IELTS Academic: 7.5 (Listening: 8.0, Reading: 8.5, Writing: 6.5, Speaking: 6.0).